

Systems of Linear Equations

ANSWER KEY



Setting up the augmented matrix

- 1 Write the augmented matrix for the system $2x + y - z = 3$, $x - 3y + 2z = -1$, $4x + y + z = 5$.

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 3 \\ 1 & -3 & 2 & -1 \\ 4 & 1 & 1 & 5 \end{array} \right].$$

Elementary row operations

- 2 List the three types of elementary row operations.
1. Swap two rows.
 2. Multiply a row by a nonzero scalar.
 3. Add a multiple of one row to another row.

Solving 2x2 systems by elimination

- 3 Solve $2x + 3y = 7$, $x - y = 1$.
From the second equation, $x = 1 + y$. Substituting: $2(1 + y) + 3y = 7 \Rightarrow 5y = 5 \Rightarrow y = 1$, so $x = 2$.
Solution: $(2, 1)$.
- 4 Solve $3x - 2y = 4$, $5x + y = 11$.
From the second equation, $y = 11 - 5x$. Substituting: $3x - 2(11 - 5x) = 4 \Rightarrow 13x = 26 \Rightarrow x = 2$, so $y = 1$.
Solution: $(2, 1)$.

Solving 3x3 systems by Gaussian elimination

- 5 Solve $x + y + z = 6$, $2y + 5z = -4$, $2x + 5y - z = 27$.
From the first equation, $x = 6 - y - z$. Substituting into the third:
 $2(6 - y - z) + 5y - z = 27 \Rightarrow 3y - 3z = 15 \Rightarrow y = 5 + z$. Substituting into the second:
 $2(5 + z) + 5z = -4 \Rightarrow 7z = -14 \Rightarrow z = -2$, so $y = 3$ and $x = 6 - 3 - (-2) = 5$. Solution: $(5, 3, -2)$.

Why row operations preserve the solution set

- 6 Explain briefly why multiplying a row by a nonzero constant c doesn't change a system's solution set.
Multiplying both sides of an equation by a nonzero c produces an equivalent equation, since the operation is invertible — dividing by c recovers the original equation exactly.
- 7 Why is the restriction $c \neq 0$ needed for this row operation?
If $c = 0$, the row becomes the zero row, which is not invertible — the original row's information is destroyed, so the resulting system need not have the same solution set as the original.

More 2x2 elimination practice

- 8 Solve $4x + y = 10$, $2x - y = 2$.

Adding the equations: $6x = 12 \Rightarrow x = 2$. From the second: $y = 2x - 2 = 2$. Solution: $(2, 2)$.

- 9 Solve $x + 2y = 5$, $3x - y = 1$.

From the first: $x = 5 - 2y$. Substituting: $3(5 - 2y) - y = 1 \Rightarrow 15 - 7y = 1 \Rightarrow y = 2$, $x = 1$. Solution: $(1, 2)$.

More 3x3 Gaussian elimination practice

- 10 Solve $2x + y + z = 5$, $x - y + 2z = 1$, $3x + 2y - z = 8$.

From the first: $y = 5 - 2x - z$. Substituting into the second:

$x - (5 - 2x - z) + 2z = 1 \Rightarrow 3x + 3z = 6 \Rightarrow x = 2 - z$. Substituting into the third:

$3(2 - z) + 2(5 - 2(2 - z) - z) - z = 8 \Rightarrow 8 - 2z = 8 \Rightarrow z = 0$. Then $x = 2$, $y = 1$. Solution: $(2, 1, 0)$.

Systems with no solution or infinitely many solutions

- 11 Solve $x + y = 3$, $2x + 2y = 8$, and explain what you find.

The second equation simplifies to $x + y = 4$, which contradicts $x + y = 3$. No values satisfy both — the system is inconsistent (no solution); the two lines are parallel and distinct.

- 12 Solve $x - y = 1$, $2x - 2y = 2$, and explain what you find.

The second equation is exactly $2 \times$ the first, so every solution of the first automatically satisfies the second — infinitely many solutions, $x = 1 + t$, $y = t$ for any t . The two equations describe the same line.

Setting up a system from a word problem

- 13 A store sells adult tickets for 8 dollars and child tickets for 5 dollars. On a day when 90 tickets were sold for 600 dollars total, set up a system of equations for the number of adult (a) and child (c) tickets sold.

$a + c = 90$; $8a + 5c = 600$.

Solving the ticket-sales system

- 14 Solve the system from the previous question.

From the first equation, $c = 90 - a$. Substituting: $8a + 5(90 - a) = 600 \Rightarrow 3a = 150 \Rightarrow a = 50$, $c = 40$.

Solution: 50 adult tickets, 40 child tickets.

Reading a solution from row echelon form

- 15 The augmented matrix $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \end{array} \right]$ is already in reduced row echelon form. State the solution.

Each row reads directly as a solved variable: $x = 3, y = -2, z = 5$.

Verifying a solution

- 16 Verify that $(x, y, z) = (1, 2, -1)$ solves $x + y + z = 2, 2x - y + z = -1, x + 2y - 3z = 8$.
 $1 + 2 + (-1) = 2$ ✓. $2(1) - 2 + (-1) = -1$ ✓. $1 + 2(2) - 3(-1) = 1 + 4 + 3 = 8$ ✓. All three check, so $(1, 2, -1)$ is a solution.
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A system with a parameter

- 17 For which value of k does $x + 2y = 3, 3x + ky = 9$ have infinitely many solutions?
Scaling the first equation by 3 gives $3x + 6y = 9$, which is identical to the second equation exactly when $k = 6$ — then the two equations describe the same line, giving infinitely many solutions.
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Consistency of a system

- 18 A system of two linear equations in x, y is consistent if it has at least one solution. Geometrically, what does an inconsistent system of two lines look like?
Two parallel, distinct lines — same slope, different intercepts — so they never meet and the system has no solution.
- 19 Is the system $2x - y = 3, 4x - 2y = 7$ consistent?
Scaling the first equation by 2 gives $4x - 2y = 6$, which contradicts $4x - 2y = 7$. The system is inconsistent (no solution).
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Number of solutions from row echelon form

- 20 A row echelon form has a row $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 5 \end{array} \right]$. What does this tell you about the system?
This row reads $0 = 5$, a contradiction — the system has no solution (it's inconsistent), regardless of what the other rows say.
- 21 A 3×3 system row-reduces to a row echelon form with a row of all zeros (no contradiction) and only 2 nonzero pivot rows. What does this tell you?
There are only 2 independent equations for 3 unknowns, so one variable is free — the system has infinitely many solutions.
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Back-substitution from an upper-triangular system

- 22** Solve by back-substitution: $x + 2y - z = 4$, $y + 3z = 5$, $2z = 6$.

From the third equation, $z = 3$. Substituting into the second: $y + 9 = 5 \Rightarrow y = -4$. Substituting into the first: $x + 2(-4) - 3 = 4 \Rightarrow x = 15$. Solution: $(15, -4, 3)$.

Homogeneous systems

- 23** A system is homogeneous if every equation equals 0 (e.g. $2x - y = 0$, $x + 3y = 0$). Explain why $x = y = 0$ is always a solution of a homogeneous system.

Substituting $x = y = 0$ makes every left-hand side (a sum of terms each with a variable factor) equal to 0, matching the right-hand side — this "trivial solution" always works, whatever the coefficients are.

- 24** Solve the homogeneous system $2x - y = 0$, $x + 3y = 0$.

From the first, $y = 2x$. Substituting into the second: $x + 3(2x) = 0 \Rightarrow 7x = 0 \Rightarrow x = 0$, so $y = 0$. Only the trivial solution $(0, 0)$.